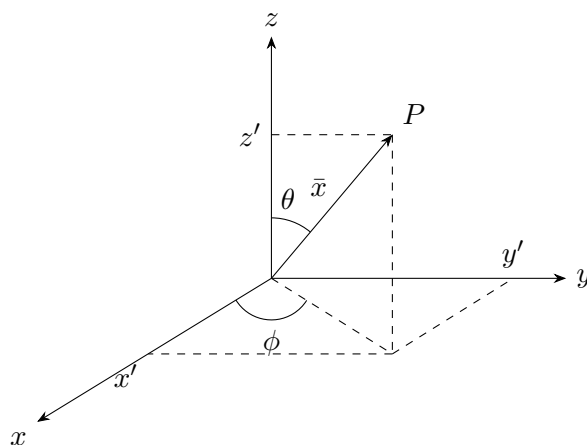


EMT Study Guide 1

This study guide will mainly follow Pollack & Stump's, with some inspiration from Griffith's and Dr. Ray's notes. It will start with a brief review of Pollack chapter 2, but really will begin with chapter 3. Practice problems are at the end. Some are based on tophat questions, but mostly they will be in the style of the tutorials.

Math Review

Vectors



$$\bar{x} = x' \hat{i} + y' \hat{j} + z' \hat{k}$$

Remember right-hand rule, $\hat{i} \times \hat{j} = \hat{k}$, so don't swap x and y axes.

$$\bar{A} \cdot \bar{B} = \sum_i A_i B_i$$

because $\hat{i} \cdot \hat{j} = 0$, $\hat{j} \cdot \hat{k} = 0$, $\hat{i} \cdot \hat{k} = 0$ (orthonormal basis).

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\bar{A} \cdot (\bar{B} \times \bar{C}) = (\bar{A} \times \bar{B}) \cdot \bar{C} \qquad \bar{A} \times (\bar{B} \times \bar{C}) = \bar{B}(\bar{A} \cdot \bar{C}) - \bar{C}(\bar{A} \cdot \bar{B})$$

$$\bar{A} \cdot \bar{B} = |\bar{A}| |\bar{B}| \cos \theta \qquad \bar{A} \times \bar{B} = |\bar{A}| |\bar{B}| \sin \theta \hat{n}$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

∇f is a vector. f is a scalar function.

$$\nabla \cdot \bar{F}(\bar{x}) = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$\nabla \cdot \bar{F}$ is a scalar. $\bar{F}(\bar{x})$ is a vector function.

$$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\nabla \times \bar{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

Integral Theorems

Gauss's Theorem

$$\oint_S \bar{F} \cdot d\bar{A} = \int_V \nabla \cdot \bar{F} d^3x$$

Everything produced inside must come out through the boundary surface S . Or, total flow out of $\bar{F}$ through S is the sum of “sources” minus “sinks.”

Stokes's Theorem

$$\oint_C \bar{F} \cdot d\bar{\ell} = \int_S (\nabla \times \bar{F}) \cdot d\bar{A}$$

Cylindrical Coordinates

Pg 34

$$\nabla f = \hat{r} \frac{\partial f}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial f}{\partial \phi} + \hat{k} \frac{\partial f}{\partial z}$$

$$\nabla \cdot \bar{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

$$\nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$

Curl:

$$(\nabla \times \bar{F})_r = \frac{1}{r} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}$$

$$(\nabla \times \bar{F})_\phi = \frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}$$

$$(\nabla \times \bar{F})_z = \frac{1}{r} \frac{\partial}{\partial r} (r F_\phi) - \frac{1}{r} \frac{\partial F_r}{\partial \phi}$$

$$x = r \cos \phi \qquad y = r \sin \phi \qquad z = z$$

Spherical Coordinates

Pg 36

$$\nabla f = \hat{r} \frac{\partial f}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}$$

$$\nabla \cdot \bar{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$$

More can be found on Pg 36, but I don't feel like copying it here.

Chapter 3: Principles of Electrostatics

Coulomb's Law

Force on q_1 due to q_2 :

$$F_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

If $\vec{r} = \bar{x}_1 - \bar{x}_2$, then

$$F_1 = \frac{q_1 q_2}{4\pi\epsilon_0} \frac{\bar{x}_1 - \bar{x}_2}{|\bar{x}_1 - \bar{x}_2|^3}$$

$\frac{1}{4\pi\epsilon_0}$ has units of $\frac{\text{N m}^2}{\text{C}^2} = 8.99 \times 10^9 \frac{\text{N m}^2}{\text{C}^2}$. C (coulomb) = 1 amp second.

Superposition

$$F_q = \sum_{k=1}^N \frac{q q_k}{4\pi\epsilon_0} \frac{\bar{x} - \bar{x}_k}{|\bar{x} - \bar{x}_k|^3}$$

Electric Field

Definition is force exerted on small test q at $\bar{x}$:

$$\bar{E}(\bar{x}) = \lim_{q \rightarrow 0} \frac{\bar{F}}{q} \quad \bar{F}_q = q \bar{E}(\bar{x})$$

$$\bar{E}(\bar{x}) = \sum_{k=1}^N \frac{q_k}{4\pi\epsilon_0} \frac{\bar{x} - \bar{x}_k}{|\bar{x} - \bar{x}_k|^3}$$

All electrostatics can be derived from Coulomb's and superposition.

Field of Charge Continuum

Denote volume charge density $\rho(\bar{x}')$ at location $\bar{x}'$:

$$\bar{E}(\bar{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\bar{x} - \bar{x}'}{|\bar{x} - \bar{x}'|} \rho(\bar{x}') d^3x'$$

If Φ is a scalar function,

$$\nabla\Phi = \left(\frac{\partial\Phi}{\partial x}, \frac{\partial\Phi}{\partial y}, \frac{\partial\Phi}{\partial z} \right)$$

$$(\nabla \times \nabla\Phi)_x = \frac{\partial}{\partial y} \frac{\partial\Phi}{\partial z} - \frac{\partial}{\partial z} \frac{\partial\Phi}{\partial y} = 0$$

Same for y and z . So $\nabla \times \nabla\Phi = 0$.

Note

$$-\nabla \frac{1}{|\bar{x} - \bar{x}'|} = \frac{\bar{x} - \bar{x}'}{|\bar{x} - \bar{x}'|^3}$$

so,

$$\bar{E}(\bar{x}) = -\nabla \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{x}') d^3x'}{|\bar{x} - \bar{x}'|} = -\nabla V(\bar{x}) \Rightarrow -\nabla^2 V = \frac{\rho}{\epsilon_0}$$

V is a scalar function, so

$$\nabla \times \bar{E} = \nabla \times (-\nabla V) = 0$$

because curl of gradient is always 0. So $\nabla \times \bar{E} = 0$.

Electric Potential and Boundary Conditions

$$\nabla \cdot \bar{E} = \frac{\rho}{\epsilon_0} \quad \text{and} \quad -\nabla^2 V = \frac{\rho}{\epsilon_0}$$

$$\bar{E}(\bar{x}) = -\nabla \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{x}') d^3x'}{|\bar{x} - \bar{x}'|}$$

where

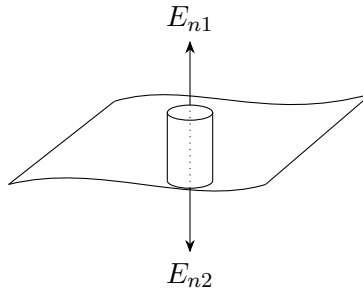
$$V(\bar{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{x}') d^3x'}{|\bar{x} - \bar{x}'|}$$

is the electric potential. For point charges it is

$$V(\bar{x}) = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{q_k}{|\bar{x} - \bar{x}'|}$$

$\nabla \times \bar{E} = 0$, so the tangential component of $\bar{E}$ must be continuous across a surface S . $\bar{E}_n$, normal to the surface, is continuous unless there is a surface charge; then there is a discontinuity of σ/ϵ_0 .

Use Gauss's law to prove that:



$$\oint \bar{E} \cdot d\bar{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

$$\oint \bar{E} \cdot d\bar{A} = \frac{\sigma dA}{\varepsilon_0}$$

$$E_{n1} dA - E_{n2} dA = \frac{\sigma dA}{\varepsilon_0}$$

$$E_{n1} - E_{n2} = \frac{\sigma}{\varepsilon_0}$$

For an infinite plane,

$$\bar{E}(\bar{x}) = \begin{cases} \frac{\sigma}{2\varepsilon_0} \hat{k} & z > 0 \text{ (above)} \\ -\frac{\sigma}{2\varepsilon_0} & z < 0 \text{ (below)} \end{cases}$$

Discontinuity of σ/ε_0 .

Dirac Delta Function

For every continuous $f(x)$, define $\delta(x)$ such that

$$\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0)$$

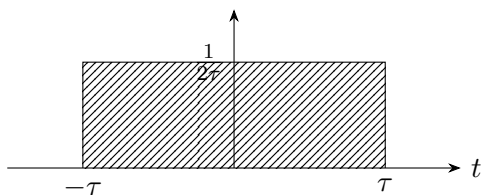
To satisfy this, $\delta(x)$ must $= 0 \forall x \neq 0$ and

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

This seems insane, and it kinda is, but watch this:

Let

$$d_{\tau}(t) = \begin{cases} \frac{1}{2\tau} & -\tau < t < \tau \\ 0 & \text{otherwise} \end{cases}$$



$$\int_{-\infty}^{\infty} d_{\tau}(t) dt = 1$$

Now imagine what happens when $\tau \rightarrow 0$. $\int_{-\infty}^{\infty} d_{\tau}(t) dt = 1$ still holds.

$$\delta(t) = \lim_{\tau \rightarrow 0} d_{\tau}(t)$$

Mathematically, this notation is horrendous and you shouldn't use it around mathematicians. Instead you would use the limit notation ¹, Anyway!

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1$$

More generally,

$$\int_{-\infty}^{\infty} \delta(x - c) f(x) dx = f(c)$$

Examples:

$$\int_{-\infty}^{\infty} x^2 \delta(2 - x) dx = (2)^2 = 4$$

$$\int_0^{\infty} x^2 \delta(x + 2) dx$$

$c = -2$, but -2 isn't in $[0, \infty]$, so this is 0.

$$\delta^3(\vec{x}) = \delta(x) \delta(y) \delta(z)$$

has units m^{-3} . Generally, units of $\frac{1}{\text{input unit}}$.

Green's Function

Green's function of $-\nabla^2$ is $G(\vec{x} - \vec{x}')$:

$$-\nabla^2 G(\vec{x} - \vec{x}') = \delta^3(\vec{x} - \vec{x}')$$

along with boundary condition $G \rightarrow 0$ at infinity.

$$G(\vec{x} - \vec{x}') = \frac{1}{4\pi|\vec{x} - \vec{x}'|} \quad \text{proof on Pg 64}$$

Pollack writes (3.58):

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} \int [-\nabla^2 G(\vec{x} - \vec{x}')] \rho(\vec{x}') d^3x'$$

¹No actual function can be 0 everywhere except one point and still have integral 1, since changing a function at a single point doesn't change its integral. Mathematicians instead treat δ as shorthand for what happens in the limit, after integrating against a smooth function f : $\lim_{\tau \rightarrow 0} \int_{-\infty}^{\infty} d_{\tau}(t) f(t) dt = f(0)$. This is the only property of δ you ever actually use.

Electric Potential

$V(\bar{x})$ is a scalar function such that

$$\bar{E}(\bar{x}) = -\nabla V(\bar{x})$$

$V(\bar{x})$ has units of volts, $1 \text{ V} = 1 \text{ J/C}$ (work/unit charge).

$$V(\bar{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\bar{x}') d^3x'}{|\bar{x} - \bar{x}'|}$$

For a point charge at the origin,

$$V(\bar{x}) = \frac{q}{4\pi\epsilon_0 r}$$

Another way to construct $V(\bar{x})$ is

$$V(\bar{x}) = - \int_{\Gamma} \bar{E} \cdot d\bar{\ell}$$

$V(\bar{x})$ is the work that must be done by an external force acting against electric force to move unit charge from $\bar{x}_0$ to $\bar{x}$ along Γ (path).

Note:

$$- \int_{\infty}^a \bar{E} \cdot d\bar{\ell} = \int_a^{\infty} \bar{E} \cdot d\bar{\ell}$$

$V(\bar{x})$ doesn't depend on the path Γ taken.

Energy of Electric Field

Note about Poisson's

$$-\nabla^2 V = \frac{\rho}{\epsilon_0} :$$

This, with boundary conditions, determines $V(\bar{x})$.

Suppose you have a stationary configuration of point charges and you move a charge Q from a to b . How much work will it take? At any point the force due to $\bar{E}$ is $Q\bar{E}$, so you must exert $-Q\bar{E}$.

$$\begin{aligned} W &= \int_a^b \bar{F} \cdot d\bar{\ell} = -Q \int_a^b \bar{E} \cdot d\bar{\ell} \\ &= Q [V(b) - V(a)] \end{aligned}$$

If you bring in Q from far far away to $\bar{r}$,

$$W = Q [V(\bar{r}) - V(\infty)]$$

So if you set the reference point at ∞ , $W = QV(\bar{r})$ (the work you must do).

Some notes on $V(\bar{r})$:

$$V(\bar{r}) = - \int_{\bar{x}_0}^{\bar{r}} \bar{E} \cdot d\bar{\ell}$$

where $\bar{x}_0$ is our reference point where $V(\bar{x}_0) = 0$, so $V(\bar{r})$ depends on what we set as our reference point $\bar{x}_0$. In this sense it is like altitude. Sea level? Height above terrain? V is all about difference.

The potential energy (not the same as electric potential) with respect to the reference point is defined as $-W$, so

$$U_q(\bar{x}) = q V(\bar{x})$$

Potential energy is well defined for any force F as long as $\nabla \times F = 0$.

Total electrostatic energy of set of charges:

$$U = \frac{1}{2} \sum_{i \neq j} \frac{q_i q_j}{4\pi\epsilon_0 |\bar{x}_i - \bar{x}_j|}$$

For a charge distribution,

$$U = \frac{1}{2} \int \frac{\rho(\bar{x})\rho(\bar{x}')}{4\pi\epsilon_0 |\bar{x} - \bar{x}'|} d^3x d^3x'$$

or

$$U = \frac{1}{2} \int \rho(\bar{x})V(\bar{x}) d^3x$$

Pollack and Stump write

$$\rho V = -\epsilon_0(\nabla^2 V)V = -\epsilon_0 \nabla \cdot (V \nabla V) + \epsilon_0(\nabla V)^2$$

I don't have a good derivation for this, but they use this to justify total energy:

$$U = \frac{\epsilon_0}{2} \int E^2 d^3x$$

and energy density of $\bar{E}$ is

$$u_E(\bar{x}) = \frac{\epsilon_0}{2} E^2(\bar{x})$$

Multipole Expansion

Point charge at origin:

$$V(\bar{x}) = \frac{q}{4\pi\epsilon_0 r} \quad \bar{E}(\bar{x}) = \frac{q \hat{r}}{4\pi\epsilon_0 r^2}$$

For two charges,

$$V(\bar{x}) = \frac{q_1}{4\pi\epsilon_0 |\bar{x} - \bar{x}_1|} + \frac{q_2}{4\pi\epsilon_0 |\bar{x} - \bar{x}_2|}$$

For $r \gg r_1$ & $r \gg r_2$,

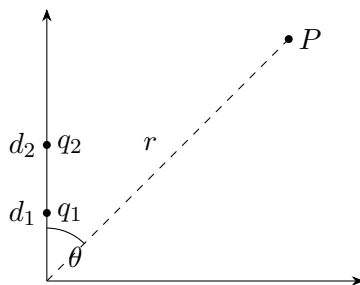
$$V(\bar{x}) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{Q}{r} + \frac{\hat{r} \cdot \bar{P}}{r^2} + \frac{\hat{r} \cdot Q_2 \cdot \hat{r}}{r^3} \right\}$$

(Deriving this is a good exercise.)

Q total charge, $\bar{P} = q_1 \bar{x}_1 + q_2 \bar{x}_2$,

$$Q_2 = \frac{q_1}{2} [3\bar{x}_1 \bar{x}_1 - r_1^2 I] + \frac{q_2}{2} [3\bar{x}_2 \bar{x}_2 - r_2^2 I]$$

Example 12 from P&S:



$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_1}{\sqrt{r^2 - 2rd_1 \cos \theta + d_1^2}} + \frac{q_2}{\sqrt{r^2 - 2rd_2 \cos \theta + d_2^2}} \right\}$$

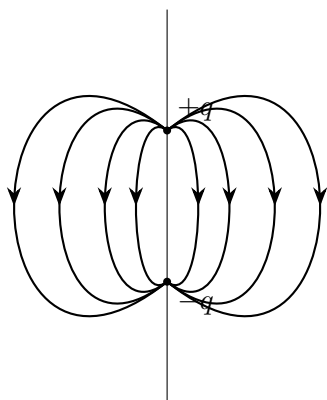
$$\hat{r} \cdot \vec{P} = (q_1 d_1 + q_2 d_2) \cos \theta$$

$$Q_2 = \frac{q_1 d_1^2 + q_2 d_2^2}{2r^3} (3 \cos \theta - 1) \quad (\text{for example 12})$$

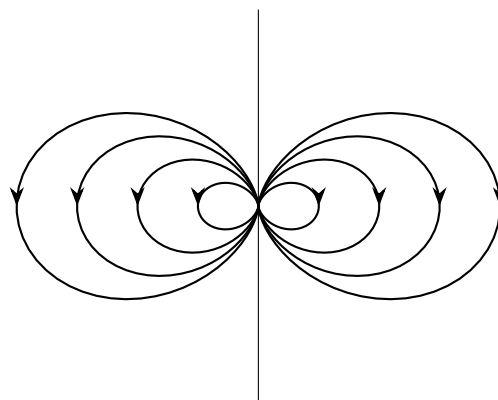
Dipoles

Dipole vectors go from negative charge to positive. $\vec{P} = q\vec{d}$. When $d \rightarrow 0$ this is a point-like dipole. If you take the origin as the dipole position, all other terms in the multipole expansion other than dipole are 0.

$$V(\vec{x}) = \frac{\vec{P} \cdot \hat{r}}{4\pi\epsilon_0 r^2}$$



Finite dipole field



Point dipole field

Not really sure why, but P&S talk about some molecules and

$$1 \text{ Debye} = 3.33 \times 10^{-30} \text{ C m}$$

$$\vec{E}(\vec{x}) = -\nabla V = \frac{3\hat{r}(\vec{P} \cdot \hat{r}) - \vec{P}}{4\pi\epsilon_0 r^3}$$

For $\bar{P}$ at origin, $\bar{P} = P_0 \hat{k}$, $\bar{E}$ in spherical is

$$\bar{E}(\bar{x}) = -\nabla V = \frac{P_0}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta})$$

Moments of General Charge Distribution

$$Q = \sum_{k=1}^N q_k \quad \bar{P} = \sum_{k=1}^N q_k \bar{x}_k \quad Q_2 = \sum_{k=1}^N \frac{q_k}{2} (3\bar{x}_k \bar{x}_k - r_k^2 I)$$

For situations with $\rho(\bar{x}')$:

$$Q = \int \rho(\bar{x}') d^3x' \quad \bar{P} = \int \bar{x}' \rho(\bar{x}') d^3x'$$

$$Q_2 = \int \frac{1}{2} (3\bar{x}' \bar{x}' - r^2 I) \rho(\bar{x}') d^3x'$$

If charge is 0, $\bar{P}$ is independent of origin. If origin is moved to $\bar{x}_0$, then

$$\bar{P}_0 = \int (\bar{x}' - \bar{x}_0) \rho(\bar{x}') d^3x' = \bar{P} - Q\bar{x}_0$$

Torque on a Dipole

A dipole in a field $\bar{E}$ experiences a torque that twists the dipole to align with $\bar{E}$. The torque is

$$\bar{N} = \int \bar{x} \times d\bar{F} = \int \bar{x} \times \bar{E}(\bar{x}) \rho(\bar{x}) d^3x$$

If $\bar{E}$ is constant on the scale of ρ (point-like dipole),

$$\bar{N} = \bar{P} \times \bar{E}$$

Chapter 4: Electrostatic Properties of Conductors

- $\bar{E}$ field is 0 inside a conductor at equilibrium.
- Potential is constant inside conductor at equilibrium.
- Charge density is zero inside conductor at equilibrium.
- We can specify net charge or potential for a conductor.
- At the surface, $E_{\text{tangent}} = 0$ and $E_{\text{normal}} = \sigma/\epsilon_0$.

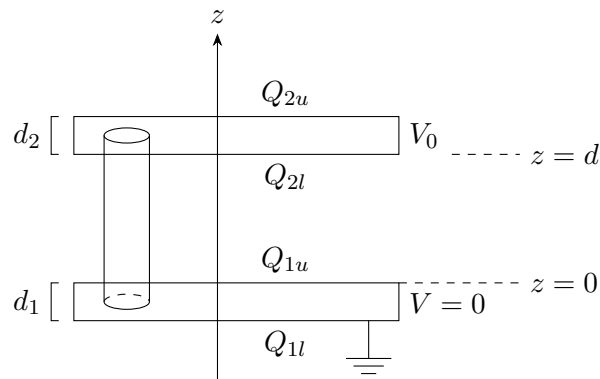
- Potential is uniquely determined up to a constant when
 - a) the charge densities and
 - b) either potentials or total charge of conductors
 are specified.

$$V(\bar{x}) = V_0 \quad \text{Dirichlet boundary condition.}$$

If surface charge density is known,

$$-\hat{n} \cdot \nabla V = \frac{\sigma}{\varepsilon_0} \quad \text{Neumann.}$$

$\vec{E}$ is 0 in any conductor cavity regardless of $\vec{E}$ outside.

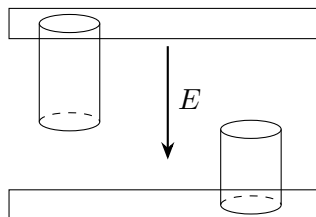


$$V(z) = V_0 \frac{z}{d} \quad \text{in region between plates}$$

$$E = -\nabla V = -\frac{V_0}{d}$$

Cylindrical Gauss surface:

Cylindrical Gauss surface:



Top:

$$\pi r^2 E = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$-\pi r^2 \epsilon_0 \frac{V_0}{d} = Q_{\text{enc}}$$

$$\sigma_{2l} = -\epsilon_0 \frac{V_0}{d}$$

Bottom:

$\vec{E} \cdot d\vec{A} < 0$ because they oppose each other

$$-\pi r^2 E = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\pi r^2 \epsilon_0 \frac{V_0}{d} = Q_{\text{enc}} \Rightarrow \sigma_{1u} = \epsilon_0 \frac{V_0}{d}$$

Two plates holding equal & opposite charges $\pm Q$ form a capacitor. Capacitance depends on geometry only.

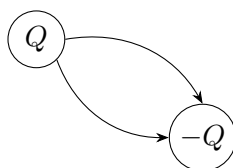
$$C = \frac{Q}{V}$$

$$= \frac{A \frac{\epsilon_0 V_0}{d}}{V_0 \frac{z}{d}} = \frac{\epsilon_0 A}{z}$$

But here $z = d$:

$$\frac{\epsilon_0 A}{d} = C$$

Energy of Capacitors



$$U = \frac{1}{2} \int \rho V d^3x$$

Because charge lies on surface,

$$U = \frac{1}{2} \int \sigma V dA$$

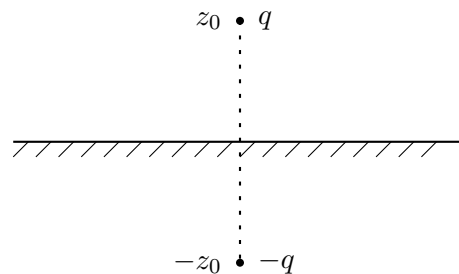
$$U_1 = -QV_1/2 \quad U_2 = QV_2/2$$

$$U_{\text{total}} = QV/2 \quad V = V_2 - V_1$$

$$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$$

Method of Images:

A charge external of a conductor creates an E field which induces a charge on the surface of the conductor. The resulting net E field outside the conductor can be modeled with a fictitious equal and opposite charge at an equal distance to the boundary.

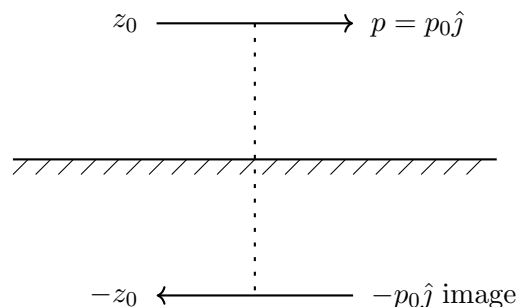


You can easily write $V(x, y, z)$ and then $\vec{E} = -\nabla V$. using $\vec{E}(r) = \frac{\sigma}{\epsilon_0} \cdot \hat{n}$ where $r = \sqrt{x^2 + y^2}$ you can show

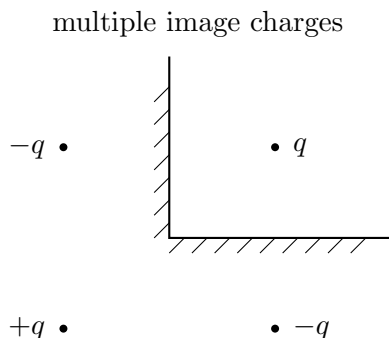
$$\sigma(r) = \frac{-q z_0}{2\pi (r^2 + z_0^2)^{3/2}}$$

* Try showing total charge on plane is $-q$

with a dipole:



$$\vec{E}(0, 0, z) = \frac{p_0 \hat{i}}{4\pi\epsilon_0} \left[\frac{-1}{(z - z_0)^3} + \frac{1}{(z + z_0)^3} \right]$$



Spherical Charges

We can find V outside of a charged sphere using Laplace's equation $\nabla^2 V = 0$. We have boundary conditions $V(a) = V_0$, $V(\infty) = 0$. (Sphere has radius a .)

Laplace's equation becomes

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0$$

We can try $V(r) \propto r^p$ with $p = 0$ or $p = -1$. Why?

Try $V = r^p$:

$$\begin{aligned} \frac{dV}{dr} &= p r^{p-1} \\ r^2 \frac{dV}{dr} &= p r^{p-1} r^2 = p r^{p+1} \\ r^2 \frac{d^2V}{dr^2} &= p(p+1) r^p \\ \frac{d^2V}{dr^2} &= \frac{1}{r^2} p(p+1) r^p = 0 \end{aligned}$$

So p must be either 0 or -1 .

If $p = 0$ then V is constant. If $p = -1$ then $V \propto 1/r$, but we can add a constant to V without violating $\nabla^2 V = 0$.

So

$$V = B + \frac{A}{r}$$

$V(\infty) = 0 \Rightarrow B = 0$. $V(a) = V_0$ requires $A = V_0 a$.

So

$$V(r) = V_0 \frac{a}{r} \quad \text{for } r \geq a$$

Practice Problems
